Cultural Bias Matters: A Cross-Cultural Benchmark Dataset and Sentiment-Enriched Model for Understanding Multimodal Metaphors

Anonymous Author

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Abstract

Metaphors pervade human communication and are deeply shaped by cultural context, yet existing multimodal metaphor datasets are predominantly English-centric, limiting the cross-cultural generalizability of computational models. We introduce **MultiMM**, a large-scale cross-cultural benchmark comprising 8,461 text–image advertisement pairs (4,397 Chinese and 4,064 English), annotated for metaphor occurrence, source/target domains, and sentiment polarity. We further propose **SEMD**, a Sentiment-Enriched Multimodal Detection model that fuses visual features from ViT, multilingual textual representations from mBERT, and sentiment embeddings to improve metaphor understanding. SEMD achieves state-of-the-art performance with macro-F1 scores of 80.16% (English) and 77.79% (Chinese) for metaphor detection, and 75.69% (English) and 70.51% (Chinese) for sentiment analysis. Our analyses reveal significant cultural divergences—for instance, positive sentiment dominates metaphoric English ads (74.75%) far more than Chinese ads (56.42%)—and show that current multimodal large language models struggle substantially with cross-cultural metaphor interpretation.

1 Introduction

Metaphor is a fundamental mechanism of human cognition, enabling people to understand abstract concepts through more concrete experiences (Lakoff, 1994). Far from being a mere rhetorical device, metaphor pervades everyday language and thought (Ortony et al., 2022), and its use is especially prominent in advertising, where visual and textual metaphors work together to create persuasive, memorable messages (Forceville, 2002). Advertisements frequently employ multimodal metaphors—mapping source domains onto target domains across image and text—to evoke emotional responses and shape consumer attitudes (Forceville and Urios-Aparisi, 2009).

Despite the importance of metaphor in multimodal communication, computational approaches to metaphor detection have largely focused on textual modalities and English-language data (Shutova et al., 2010; Choi et al., 2021; Su et al., 2021). Recent efforts have begun to address multimodal metaphor understanding through datasets and models that combine visual and linguistic information (Zhang et al., 2021; Xu et al., 2022; Liang et al., 2022). However, these resources remain predominantly monolingual and monocultural, overlooking the well-documented fact that metaphor use varies significantly across cultures (Kövecses, 2008; Ibarretxe-Antuñano, 2013). Chinese and English cultures, for example, differ in how they conceptualize emotions, values, and abstract ideas through metaphor (Pritzker, 2003; Dwairy, 2009), yet no existing benchmark enables systematic cross-cultural evaluation of multimodal metaphor processing.

This gap presents a critical challenge: models trained and evaluated on culturally homogeneous data may encode systematic biases that undermine their applicability in multilingual and multicultural settings. Without cross-cultural benchmarks, we cannot measure or mitigate such biases.

To address these limitations, we introduce **MultiMM**, a cross-cultural multimodal metaphor benchmark consisting of 8,461 text–image advertisement pairs, balanced between Chinese (4,397) and English (4,064) sources. Each instance is annotated for metaphor occurrence, source and target domains, and sentiment polarity (negative, neutral, or positive), achieving substantial inter-annotator agreement (Fleiss' $\kappa = 0.73$ for metaphor, 0.82 for sentiment). MultiMM supports two tasks: metaphor detection and sentiment analysis.

We further propose **SEMD** (Sentiment-Enriched Multimodal Detection), a model that fuses ViT-based image features (Dosovitskiy et al., 2020) with mBERT text representations (Devlin et al.,

2019) and auxiliary sentiment embeddings. The key insight is that sentiment carries culturally informative signals that benefit metaphor interpretation. SEMD achieves state-of-the-art results on both tasks, outperforming 18 baselines including strong multimodal models. Notably, we find that current multimodal large language models (GPT-4o, LLaVA, Gemini, Qwen2.5-VL) underperform specialized models, particularly on Chinese data, suggesting that cultural competence remains a significant challenge.

Our main contributions are as follows:

- We construct **MultiMM**, the first large-scale cross-cultural multimodal metaphor benchmark with fine-grained domain and sentiment annotations across Chinese and English advertisements.
- We define two evaluation tasks—metaphor detection and sentiment analysis—and conduct a comprehensive evaluation of 18 baselines spanning text-only, vision-only, multimodal, and large language models.
- We propose **SEMD**, a sentiment-enriched multimodal model that consistently improves cross-cultural metaphor detection and sentiment analysis by leveraging affective signals.
- We provide in-depth cross-cultural analyses, including translation-based experiments, that quantify cultural transfer gaps and offer insights for building more culturally robust multimodal systems.

2 Related Work

Conceptual Metaphor Theory. Conceptual Metaphor Theory (Lakoff, 1994) posits that metaphor is not merely a linguistic phenomenon but a fundamental cognitive mechanism by which humans understand abstract concepts through concrete experiences. This framework has been extended to account for cultural variation in metaphorical conceptualization (Kövecses, 2010) and has motivated computational approaches to metaphor processing (Ortony et al., 2022). Cross-cultural studies reveal that while some conceptual metaphors are near-universal, many are culture-specific: Ibarretxe-Antuñano (2013) demonstrate that cultural context shapes metaphor interpretation, Kövecses (2008) shows systematic variation in metaphorical mappings across languages, Pritzker (2003) documents distinct metaphorical patterns in Chinese emotional discourse, and Dwairy (2009) highlights how cultural values influence metaphor use in therapeutic contexts.

Multimodal Metaphor. Forceville and Urios-Aparisi (2009) extended metaphor analysis beyond language to multimodal contexts, establishing theoretical foundations for studying how metaphors are constructed across visual and verbal modalities. Subsequent work has examined visual and multimodal metaphor in advertising (Forceville, 2017, 2002), demonstrating that advertisements frequently employ cross-modal mappings to communicate persuasive messages. These theoretical insights have motivated the development of computational approaches for multimodal metaphor detection.

Computational Metaphor Detection. Computational metaphor detection has progressed from linguistic rule-based methods (Shutova et al., 2010) to neural approaches. Mao et al. (2018) introduced word-level metaphor detection using contextualized representations, while Choi et al. (2021) proposed MelBERT, leveraging BERT for metaphor identification. Su et al. (2021) and Liu et al. (2020) further improved detection through enhanced contextual modeling. In the multimodal domain, Zhang et al. (2021) introduced MultiMET, the first multimodal metaphor detection dataset, followed by Met-Meme (Xu et al., 2022) for internet memes. Liang et al. (2022) proposed multi-grained fusion methods, and Maity et al. (2022) developed multitask learning frameworks for joint metaphor and sentiment prediction. However, these resources are predominantly English-only, leaving cross-cultural multimodal metaphor understanding largely unexplored.

Sentiment and Pre-trained Models. Sentiment analysis has deep connections to metaphor processing, as metaphors frequently carry affective connotations (Citron and Goldberg, 2014). Ge et al. (2022) highlighted the importance of integrating sentiment understanding into broader NLP tasks. Our work builds on pre-trained multilingual models—mBERT (Devlin et al., 2019) for cross-lingual text encoding and XLM-RoBERTa (Conneau et al., 2020) as a baseline—alongside ViT (Dosovitskiy et al., 2020) for visual feature extraction, combining these with sentiment representations to bridge cultural gaps in multimodal

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metaphor understanding.

Cross-Lingual and Cultural NLP. Advances in cross-lingual NLP have yielded powerful multilingual pre-trained models that transfer representations across languages (Conneau et al., 2020; Devlin et al., 2019). However, language transfer does not imply cultural transfer: models may fail to capture culturally specific pragmatic conventions and figurative patterns present in another language (Kövecses, 2008). Cultural differences are particularly salient in advertising, where visual and linguistic choices reflect local values, aesthetic sensibilities, and communicative norms (Forceville, 2002). Cross-cultural NLP studies have documented systematic disparities in sentiment expression (Ge et al., 2022) and metaphor use (Ibarretxe-Antuñano, 2013) across languages, motivating dedicated cross-cultural benchmarks. MultiMM addresses this gap for multimodal metaphor, providing the first benchmark to systematically quantify cultural divergence in figurative multimodal communication.

Multimodal Large Language Models and Figurative Language. Multimodal large language models (MLLMs) such as GPT-4o (Hurst et al., 2024), LLaVA (Liu et al., 2024), Gemini (Team et al., 2023), and Qwen2.5-VL (Bai et al., 2025) have demonstrated impressive general vision–language capabilities but remain under-evaluated on figurative and culture-specific content. While MLLMs excel at literal image description and factual question answering, figurative language comprehension requires implicit cultural knowledge and pragmatic reasoning that may not emerge from standard pre-training objectives. Recent studies suggest that MLLMs underperform specialized models on figurative tasks (Maity et al., 2022), yet systematic cross-cultural evaluations of MLLMs on multimodal metaphor remain scarce. Our benchmark directly addresses this gap by providing a rigorous evaluation framework for both fine-tuned models and zero-shot MLLMs across Chinese and English.

3 The MultiMM Dataset

3.1 Data Collection

MultiMM comprises 8,461 text–image advertisement pairs drawn from two cultural contexts. **Chinese advertisements** (4,397 instances) were collected via Baidu keyword search using terms spanning product categories, public service announcement (PSA) domains, and metaphor-relevant topics, supplemented by the iFlytek advertisement set. **English advertisements** (4,064 instances) were sourced from a publicly available dataset of English product and PSA materials (Zhang et al., 2021).

Figure 1: An annotated advertisement example from MultiMM, illustrating the four annotation fields: metaphor occurrence, target domain, source domain, and sentiment polarity.

Figure 2: Cross-cultural multimodal metaphor examples from Chinese (CN) advertisements in MultiMM, illustrating culturally specific visual–textual mappings.

Raw images underwent a rigorous filtering pipeline: exact duplicates were removed using MD5 hashing (Rivest, 1992); non-advertisement images, blurred or low-resolution images below 350×350 pixels, and images without identifiable textual content were discarded. Textual content was extracted using PaddleOCR (Du et al., 2020) with subsequent manual correction to ensure accuracy.

3.2 Annotation Process

Annotation guidelines were derived from the Metaphor Identification Procedure (Steen et al., 2010) and multimodal metaphor theory (Forceville

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Figure 3: Cross-cultural multimodal metaphor examples from English (EN) advertisements in MultiMM, illustrating culturally specific visual–textual mappings.

and Urios-Aparisi, 2009). Each instance is annotated with four fields: (1) **Metaphor Occurrence** (binary: metaphorical/literal); (2) **Target Domain** (the concept being described, extracted from text and verbalized visual content); (3) **Source Domain** (the concept used to characterize the target, verbalized from the visual modality); and (4) **Sentiment Category** (negative, neutral, or positive).

Expert annotators were organized into groups: five for Chinese (in 2-2-1 groupings) and three for English (in 2-1 groupings), with adjudication rounds to resolve disagreements. Multiple training rounds and iterative guideline refinement ensured consistency. To mitigate cultural bias, annotators from diverse backgrounds were recruited. An annotated example is shown in Figure 1, and cross-cultural metaphor examples appear in Figures 2 and 3.

Inter-annotator reliability was measured using Fleiss' κ (Fleiss, 1971): 0.73 for metaphor occurrence, 0.70 for target domain, 0.66 for source domain, and 0.82 for sentiment category—indicating substantial to near-perfect agreement across all annotation dimensions.

3.3 Dataset Statistics

Table 1 presents a statistical overview of MultiMM. The dataset contains 8,461 instances with 4,772 metaphorical (56.4%) and 3,689 literal (43.6%) examples. Chinese texts are notably longer on average (33 tokens) than English (15 tokens), reflecting differences in advertising conventions. The data is split into training (6,768), validation (846), and test (847) sets with stratified sampling to preserve class balance across both languages.

Item	CN	EN	Total
Total	4,397	4,064	8,461
Metaphorical	2,583	2,189	4,772
Literal	1,814	1,875	3,689
Total Words	145,312	68,189	213,501
Average Words	33	15	24
Training Set Size	3,517	3,251	6,768
Validation Set Size	440	406	846
Test Set Size	440	407	847

Table 1: Statistical overview of MultiMM. CN refers to Chinese data, while EN refers to English data.

4 Data Analysis

4.1 Sentiment Distribution

Figure 4 shows the sentiment distribution for metaphorical and literal instances in both languages. A striking cultural asymmetry emerges: positive sentiment strongly dominates metaphorical English advertisements (74.75%) but is less prevalent in Chinese metaphorical ads (56.42%). This divergence reflects broader cultural differences in advertising strategy and emotional communication norms. In Chinese advertising, neutral and negative sentiments—particularly in PSA-style messaging—are more commonly conveyed through metaphorical framing. By contrast, English advertising predominantly uses positive metaphors to promote products and aspirational lifestyles. This sentiment asymmetry also interacts with domain: in Chinese advertisements, negative sentiment disproportionately co-occurs with health and safety PSA domains, whereas positive sentiment is more strongly associated with consumer product advertising. Understanding these domain–sentiment interactions is important for fine-grained cross-cultural analysis.

4.2 Source Domain Analysis

Figure 5 illustrates the distribution of source domain vocabulary across the two languages. Both languages draw on nature, body, and animal domains, reflecting universal cognitive metaphor mappings (Kövecses, 2010). However, cross-cultural differences are evident: Chinese advertising shows a higher prevalence of nature-based metaphors (e.g., water, mountain), consistent with Chinese cultural symbolism, while English advertising draws

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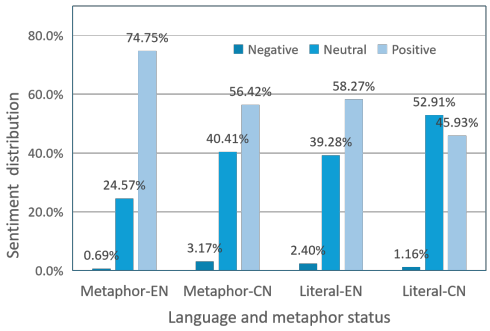

Figure 4: Sentiment category distribution in metaphorical and literal data from Chinese (CN) and English (EN) advertisements. Metaphorical English ads are dominated by positive sentiment, while Chinese metaphorical ads show a more balanced distribution.

more heavily on technology and sports domains. Figure 6 visualises these differences through word clouds of source vocabulary stratified by sentiment category, further highlighting how affective stances are expressed through culturally specific figurative language.

Target domain analysis reveals equally instructive cross-cultural patterns. In both Chinese and English advertisements, the most frequent target domains are product quality, health, and emotion, consistent with the persuasive goals of advertising discourse. However, Chinese advertisements exhibit a notably higher proportion of abstract target domains—such as national spirit, social harmony, and collective well-being—reflecting Confucian values and state-sponsored messaging norms. English advertisements, by contrast, favour concrete target domains centred on individual lifestyle benefits and product performance. These differences interact systematically with source domain choice: Chinese advertisers tend to map social-harmony targets onto nature-based sources (e.g., SOCIAL HARMONY IS A FLOWING RIVER), while English advertisers more frequently map lifestyle targets onto technology or sports sources (e.g., LIFE IS A RACE). Such domain-pairing asymmetries underscore why cross-cultural transfer remains challenging even for models with strong multilingual text representations.

5 Methodology: SEMD

We propose **SEMD** (Sentiment-Enriched Multimodal Detection), a multimodal fusion model that integrates visual, textual, and sentiment features for both metaphor detection and sentiment analysis.

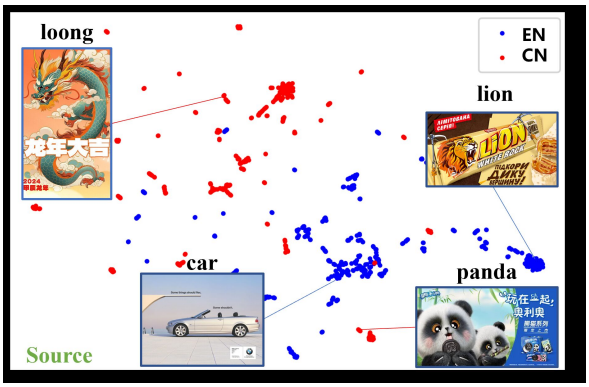

Figure 5: Distribution of source domain vocabulary in Chinese (CN) and English (EN) advertisements.

The architecture is illustrated in Figure 7.

5.1 Feature Extraction

Image features. We use ViT (`vit-base-patch16-224`) (Dosovitskiy et al., 2020) to encode advertisement images. The `[CLS]` token representation from the final transformer layer serves as the image feature $\mathbf{v} \in \mathbb{R}^{768}$.

Text features. Advertisement text extracted via OCR is encoded by mBERT (`bert-base-multilingual-cased`) (Devlin et al., 2019), which supports both Chinese and English within a shared multilingual representation space. The `[CLS]` token embedding $\mathbf{t} \in \mathbb{R}^{768}$ serves as the text feature. Input sequences are truncated or padded to a maximum of 30 tokens.

Sentiment features. For metaphor detection, we incorporate auxiliary sentiment embeddings $\mathbf{s} \in \mathbb{R}^{768}$ derived from a pre-trained sentiment analysis model. Our data analysis demonstrates a strong link between sentiment polarity and metaphor occurrence (Section 4), motivating sentiment as an additional modality.

5.2 Multimodal Fusion and Classification

For **metaphor detection**, all three feature vectors are concatenated:

$$\mathbf{f}_{\text{MD}} = [\mathbf{v};\, \mathbf{t};\, \mathbf{s}] \in \mathbb{R}^{2304} \tag{1}$$

For **sentiment analysis**, sentiment inputs are withheld to avoid information leakage, and only image and text features are fused:

$$\mathbf{f}_{\text{SA}} = [\mathbf{v};\, \mathbf{t}] \in \mathbb{R}^{1536} \tag{2}$$

In both cases, the fused representation is passed through a feed-forward classifier with dropout (rate 0.3) for label prediction.

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Figure 6: Word clouds of source domain vocabulary across positive, neutral, and negative sentiment categories in Chinese (CN) and English (EN) advertisements.

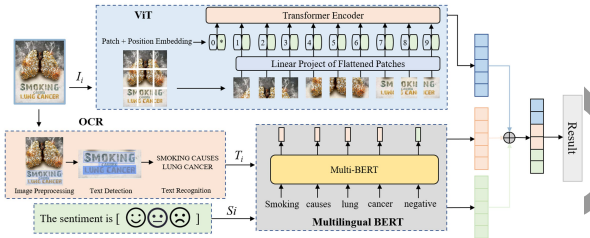

Figure 7: The SEMD framework consists of three branches: image features from ViT, text features from mBERT (applied to OCR-extracted text), and auxiliary sentiment embeddings. All three are fused via concatenation for metaphor detection; sentiment features are omitted for sentiment analysis.

5.3 Training Details

Pre-trained encoders are frozen during training. We use the AdamW optimizer (Loshchilov and Hutter, 2018) with cross-entropy loss, training for 10 epochs with a batch size of 64. The learning rate is selected from $[3 \times 10^{-5}, 5 \times 10^{-4}]$ via grid search on the validation set. All models are implemented in PyTorch (Paszke et al., 2019). All experiments are conducted on NVIDIA A100 GPUs. Early stopping with a patience of 3 epochs is applied based on validation macro-F1. To address class imbalance in sentiment analysis, where neutral instances are underrepresented, we apply class-weighted cross-entropy loss with weights inversely proportional to class frequency. Image inputs are further augmented during training via random horizontal flipping and colour jitter to improve visual encoder generalisation.

6 Experiments

6.1 Experimental Setup

We evaluate SEMD against 18 baselines spanning three modality categories, color-coded in the results table:

- **Text-only models** (cyan): mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), SixTP (Chen et al., 2022), LaBSE; generative LLMs including GPT-3.5-Turbo (Achiam et al., 2023), Llama2-8B (Touvron et al., 2023), Llama3.1-70B (Dubey et al., 2024), and DeepSeek R1-70B (DeepSeek-AI et al., 2025).
- **Vision-only models** (green): VGG16 (Simonyan and Zisserman, 2014), ResNet50 (He et al., 2016), and ViT (Dosovitskiy et al., 2020).
- **Multimodal models** (orange): mBERT-Res (mBERT + ResNet), CMGCN (Liang et al., 2022), CES (Zhang et al., 2023), MET (Zhang et al., 2021), MFC (Xu et al., 2022); and MLLMs: GPT-4o (Hurst et al., 2024), LLaVA (Liu et al., 2024), Gemini (Team et al., 2023), and Qwen2.5-VL (3B/7B/72B) (Bai et al., 2025).

All fine-tuned models are trained on the training split and selected on the validation split. MLLMs are evaluated zero-shot with task-specific prompts. For MLLM evaluation, prompts are standardised across all models and include a task description, output format instructions, and a concise definition of multimodal metaphor; for Chinese test instances, prompts are provided in Chinese to maximise MLLM performance. All MLLM experiments use temperature 0 to ensure reproducibility. Evaluation uses accuracy, macro-precision, and macro-F1 on the 847-instance test set.

6.2 Main Results

Table 2 reports results on both tasks.

Metaphor detection. SEMD achieves the highest F1 scores of 80.16% (EN) and 77.79% (CN), outperforming the best multimodal baselines: CMGCN (79.04% EN, 74.91% CN) and

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Model	Metaphor Detection F1 (%)		Sentiment Analysis F1 (%)	
	EN	CN	EN	CN
Random	46.70	41.20	40.41	35.02
mBERT	64.00	65.52	62.83	58.43
XLM_R	69.90	62.38	69.44	66.21
SixTP	68.61	65.60	64.43	65.59
LaBSE	68.15	66.41	69.93	69.02
GPT-3.5-TURBO	30.95	15.49	42.06	52.08
Llama2-8B	22.58	19.73	40.11	50.72
Llama3.1-70B	44.04	49.70	33.46	30.90
Deepseek R1-70B	44.06	47.95	53.38	49.01
VGG16	70.64	67.85	59.90	59.02
ResNet50	71.46	68.56	57.49	58.86
ViT	71.67	69.04	61.46	62.17
mBERT-Res	77.83	74.40	69.75	68.84
CMGCN	79.04	74.91	72.79	70.19
CES	78.45	75.53	73.24	70.26
MET	76.04	72.51	71.95	67.17
MFC	75.84	73.25	71.93	69.60
GPT-4O	64.00	67.00	36.00	29.00
LLaVA	73.31	69.84	56.72	53.21
Gemini	64.19	68.51	53.61	43.48
Qwen2.5-VL-3b	57.30	41.69	52.50	55.26
Qwen2.5-VL-7b	56.62	54.51	50.71	55.15
Qwen2.5-VL-72B	59.12	67.66	50.44	57.17
SEMD	**80.16**	**77.79**	**75.69**	**70.51**

Table 2: Experimental results on metaphor detection and sentiment analysis tasks. The best overall results are highlighted in **bold**, while the best results within each modality are underlined. Models are categorized by modality, with textual, visual, and multimodal models highlighted in cyan, green, and orange, respectively.

CES (78.45% EN, 75.53% CN). Among unimodal text models, LaBSE achieves the best text-only result (66.41% CN), while ViT leads among vision-only models (71.67% EN, 69.04% CN). The consistent advantage of multimodal models over both unimodal alternatives confirms the importance of cross-modal feature integration for metaphor detection.

Sentiment analysis. SEMD achieves 75.69% F1 (EN) and 70.51% F1 (CN), surpassing CES (73.24% EN, 70.26% CN) and other multimodal baselines. For text-only models, LaBSE leads with 69.93% (EN) and 69.02% (CN), reflecting the benefit of language-agnostic representations for cross-lingual sentiment understanding.

MLLMs underperform. Notably, all MLLMs—including GPT-4o, LLaVA, Gemini, and Qwen2.5-VL at all scales—substantially underperform fine-tuned models on both tasks, with the largest gaps on Chinese data. For example, GPT-4o achieves only

Sentiment Feature	Fusion Method	EN F1 (%)	CN F1 (%)
w/o	add	75.95	72.05
	max	76.29	74.09
	concat	78.64	74.84
w/	add	77.76	74.37
	max	78.59	74.37
	concat	**80.88**	**77.39**

Table 3: Results of the ablation study. The best and second-best results are highlighted in **bold** and underline, respectively.

64.00% (EN) and 67.00% (CN) on metaphor detection, compared to SEMD's 80.16% and 77.79%. This finding indicates that current MLLMs lack the culturally-grounded figurative reasoning required for multimodal metaphor understanding.

Cultural performance gap. Performance is consistently higher on English than Chinese across nearly all models, reflecting the greater subtlety of Chinese metaphors and the cultural specificity of their visual–textual mappings. SEMD narrows this gap (2.37% difference) compared to CMGCN (4.13%) and CES (2.92%), highlighting the benefit of sentiment-enriched fusion for the more challenging Chinese subset. At the instance level, multimodal models benefit from high-resolution images: on advertisements with visually complex metaphors, ViT-based models outperform ResNet-based ones by up to 4% F1, suggesting that patch-level attention captures localised visual cues that signal metaphorical mappings.

6.3 Ablation Study

Table 3 examines the contribution of sentiment features and the choice of fusion method.

Concatenation consistently outperforms element-wise addition and max-pooling across both sentiment conditions. The inclusion of sentiment features yields consistent improvements regardless of fusion method: concat+sentiment achieves 80.88% EN and 77.39% CN, compared to 78.64% EN and 74.84% CN without sentiment features—gains of +2.24% and +2.55% respectively. These results confirm that affective signals provide a culturally universal bridge that enhances multimodal metaphor detection across both languages.

6.4 Cross-Cultural Transfer

To quantify the cultural transfer gap independently of model capacity, we conduct bidirectional machine translation experiments: test instances from

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		mBERT	mBERT-Res	SEMD
EN → CN	Acc (%)	64.29	76.35	78.57
	Pre (%)	63.72	76.73	78.19
	F1 (%)	59.70	75.63	77.64
CN → EN	Acc (%)	65.22	73.86	75.45
	Pre (%)	67.46	75.90	77.73
	F1 (%)	65.31	73.96	75.53

Table 4: The results of metaphor detection tasks after Chinese-English translation. Experimental results on metaphor detection task via bidirectional Chinese-English translation.

each language are machine-translated into the other, and evaluated by models trained on the target language.

Table 4 shows that SEMD consistently achieves the best cross-lingual transfer: 77.64% F1 (EN→CN) and 75.53% F1 (CN→EN), outperforming mBERT-Res (75.63% and 73.96%) and mBERT (59.70% and 65.31%). Compared to native-language performance (80.16% EN, 77.79% CN), the drop under translation (approximately 2–3%) reveals a systematic cultural transfer gap that cannot be attributed to translation quality alone. The fact that this gap is smaller for SEMD than for other models suggests that sentiment-enriched fusion supports more culturally robust representations. These findings demonstrate that translation alone is insufficient to bridge cultural differences in multimodal metaphor.

6.5 Error Analysis

We analyse error patterns to characterise the remaining challenges. For **metaphor detection**, models achieve approximately 86% accuracy on English but only 73% on Chinese test instances. Chinese metaphors tend to rely on more indirect imagery and culturally specific visual–textual associations, making them harder for models trained on generic multilingual representations. For **sentiment analysis**, neutral sentiment is the most difficult category to detect correctly in Chinese (55% accuracy vs. 62% in English), likely due to higher lexical ambiguity and context-dependence in Chinese advertising texts. These findings motivate future work on culturally-specific pre-training and richer domain-specific resources for Chinese multimodal metaphor. We further find that models frequently confuse nature-imagery source domains (e.g., water, mountain) with literal description in Chinese ads, as these domains carry culturally embedded symbolic meanings that require world knowledge beyond generic multilingual pre-training.

7 Conclusion

We presented MultiMM, the first cross-cultural multimodal metaphor benchmark, comprising 8,461 Chinese–English advertisement pairs annotated for metaphor occurrence, source/target domains, and sentiment polarity. A comprehensive evaluation of 18 baselines reveals systematic cultural biases across all model types, with consistent performance gaps between English and Chinese. We proposed SEMD, a sentiment-enriched multimodal fusion model that achieves state-of-the-art performance on both metaphor detection and sentiment analysis across both languages. Ablation experiments confirm that sentiment representations provide a culturally universal affective signal that bridges cross-cultural differences in metaphor expression. Translation-based transfer analyses further demonstrate that cultural gaps cannot be closed by translation alone, motivating the development of genuinely cross-cultural representations. We hope MultiMM and SEMD serve as a foundation for more culturally robust and inclusive multimodal NLP research. Despite these contributions, MultiMM has several limitations. The benchmark covers only Chinese and English, and future work should extend to typologically diverse language pairs, including low-resource languages where cultural metaphor studies remain scarce. The dataset is domain-specific (advertisements), and models may not generalize directly to other figurative genres such as news, social media, or literary text. Additionally, while machine translation enables controlled transfer experiments, translated texts may retain source-language pragmatic cues that artificially inflate or deflate measured transfer performance; human-translated test sets would provide cleaner estimates. Finally, SEMD relies on pre-computed sentiment embeddings that may themselves encode cultural biases, motivating future development of sentiment resources purpose-built for cross-cultural figurative language. Future work will extend to additional language pairs, explore culture-specific pre-training, and develop interpretable models that can explain cross-cultural metaphorical mappings.

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